

Critical Design Review



Psyche-ESE

Extraterrestrial Seismic Exploration

*SES*410*

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Team Contributions

- Elysian Hudson - Team Leader
- Caleb Zick - Model Lead
- Daniel Joseph - Simulation Lead
- Jacob Coons - Computer Lead
- Spencer Emmons - Science Lead
- Syed Hamza Ayaz - Testing Lead



Agenda

- Science Question
- Background
- Motivation
- Science Traceability Matrix (STM)
- CONOPS
- System Block Diagram
- Requirement Flowdown
 - Mission-Level Requirements
 - Subsystem Definitions, Requirements, & Design
- Conclusion
- Next steps



Science Question



- ◆ By utilizing a computer simulation of a simplified model of 16-Psyche, determine whether it is feasible to map the subsurface via seismic wave propagation.



Background - 16 Psyche



- 16 Psyche is a large asteroid within the main asteroid belt, presumed to be largely composed of metal & hypothesized to be the partial core of a planetesimal.
- Named after the Greek goddess of the soul.
- An orbiter mission to investigate the asteroid is currently underway, scheduled to arrive in 2029.
- Current NASA Psyche mission:
 - Instrument Suite: Multispectral Imager, Gamma-ray and neutron spectrometer, Magnetometer, Gravity science investigation
- Mission precedents: NASA's InSight, Deep Impact, LCROSS missions. JAXA's Hayabusa2 mission.
- Single-site and/or multi-site seismology used to develop 3-dimensional maps of subsurface structures of planetary bodies.

Background - Software Package



- What is SeisSol?
 - A software package for simulating wave propagation based on the arbitrary high-order accurate derivative discontinuous Galerkin method (ADER-DG)
- What does it do?
 - Utilizes meshes to approximate 3D model geometries
 - Specifies material elasticity to approximate geological subsurface properties
 - Produces reliable and sufficiently accurate synthetic seismograms and other seismological data sets
- Why does it work for us?
 - Can select source wave generation point(s) with whatever energy we decide
 - Can get detailed seismological data at any selected detection point (ground displacement and wave energy) in order to answer our science question.

SeisSol

Motivation



- Very little is known about the interior structures of small planetary bodies.
- Knowledge about 16 Psyche currently relies on remote sensing observations.
- Directly investigate 16 Psyche to help determine if it is the core of a planetesimal, using geophysics and seismology to explore the subsurface.
- Possibly differentiated/layered.
- Learn more about the planetary formation/evolution.
- Future implications for planetary defense and resource exploitation.

Motivation–Decadal Survey



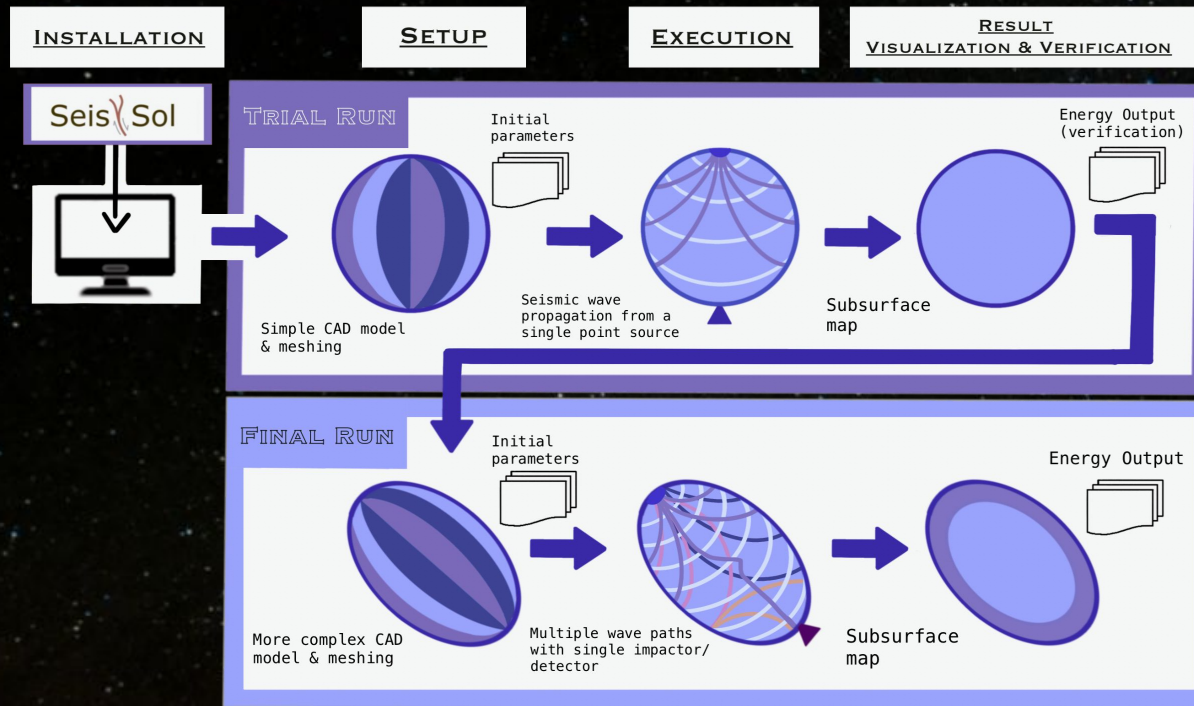
- 2023-2032 Decadal Survey Key Questions:
 - Q1 What were the initial conditions in the solar system, what processes led to the production of planetary building blocks, and what was the nature and evolution of these materials?
 - 16 Psyche is a presumed core, and therefore possibly a remnant planetary building block. Learning about its subsurface structure can tell us about the processes that lead to its formation, including processes such as differentiation.
 - Q4 How have the population of solar system bodies changed through time, how has bombardment varied across the solar system, and how have collisions affected the evolution of planetary bodies?
 - 16 Psyche may have experienced a hit-and-run event(s) during formation, and the subsurface structures can tell us about the impact history of the asteroid.
 - Q5 How did the interiors of solid bodies evolve, how is this evolution recorded in a body's physical and chemical properties, and how are solid surfaces shaped by internal and external processes?
 - 16 Psyche can be directly investigated to learn about the internal evolution of other solid bodies in our Solar System. Its internal structure will have recorded the processes responsible for its shaping.

Science Traceability Matrix (STM)

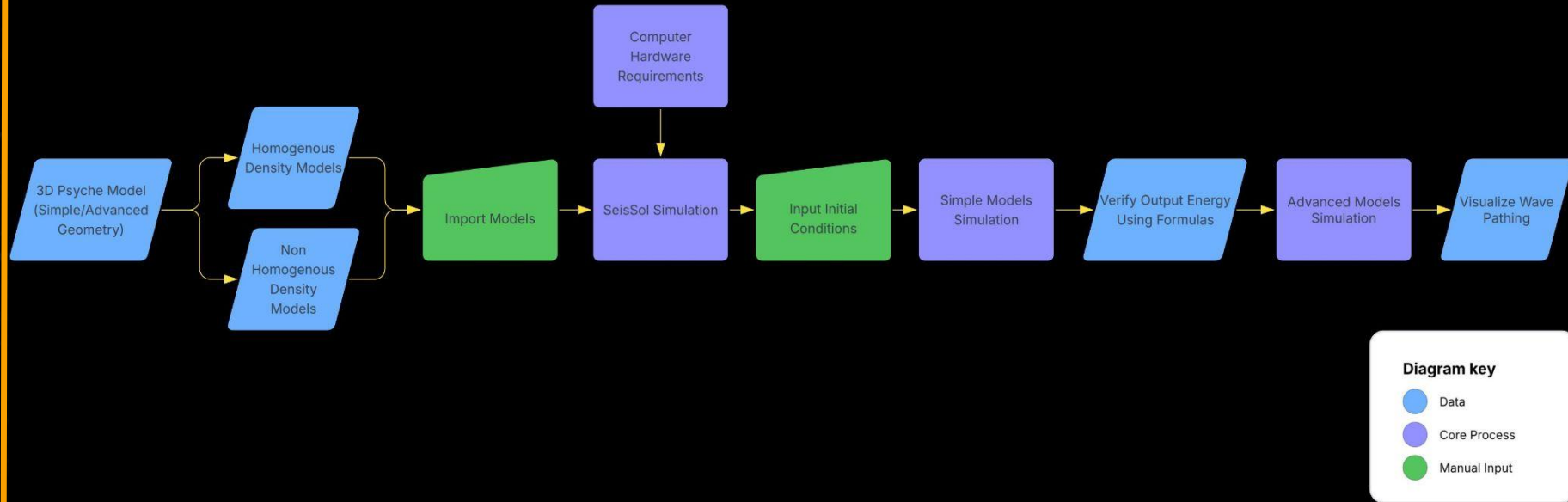


Objective	Measurement	Measurement Requirements	Technologies	Technological Requirements	Data Product
Simulate wave propagation through a model of 16 Psyche	Simulated seismic waves that fully travel through the interior of the model	Calculated minimum kinetic energy of the impact is 4 MJ	-Software -CPU -Memory -CAD Model	<u>Software</u> -SeisSol Simulation Software and dependencies (see software subsystem slide) <u>CPU</u> -Single chip vs cluster vs supercomputer setup (TBD) -Performance: minimum clock speed, # of cores, # of threads (TBD) <u>Memory</u> -Minimum amount of RAM (TBD) -Minimum DDR Performance (TBD) <u>CAD Model</u> -Minimum accuracy of bilayer, non-homogeneous, order of magnitude density estimate of 16 Psyche	Seismic wave energies at selected measurement point
	Detect simulated signal at selected point on the model of 16 Psyche	Minimum measurable ground acceleration spectral density of: $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$			Surface displacement at selected measurement point
	Propagation of seismic waves	Maximum deviation from expected wave pathing of no more than 1 degree			Animated visualization of seismic waves traversing the model (.xdmf)

Psyche CONOPS



System Block Diagram



Requirements Levels

L1: Mission and Measurement Requirements

L2: Subsystem Requirements



Requirements

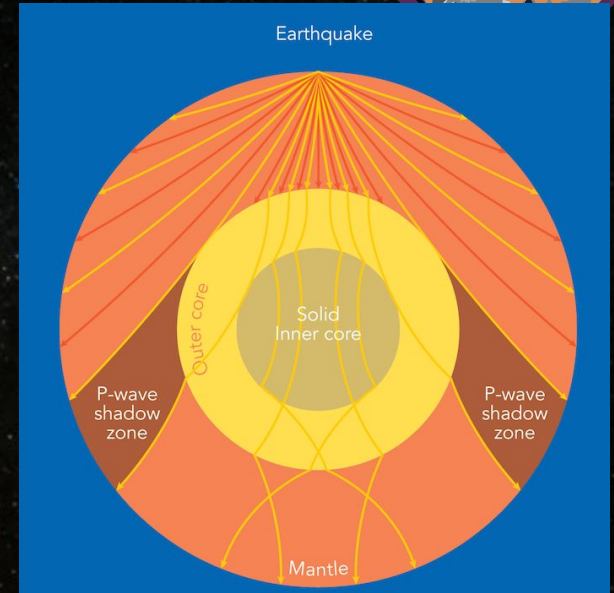


Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
L1.1	Propagation of seismic waves - The simulation shall model, to within 10% of expected parameters, the propagation of seismic waves through a 3d body.	Test, Analysis, Demonstration	Science Question
L1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	Test, Demonstration	Science Question
L1.3	Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments and simulated asteroid.	Inspection	Science Question

L1 Requirement – 1.1

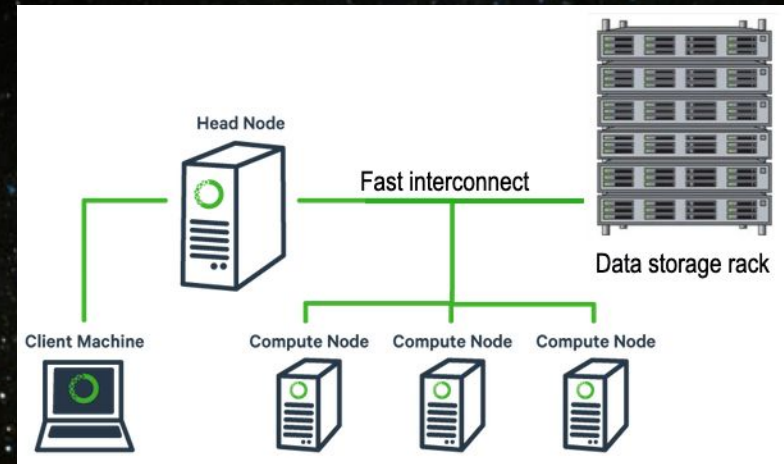
- Propagation of seismic waves - The simulation shall model the propagation of seismic waves through a 3d body.
- Shall be tested using a simplified model and the results compared to manually calculated results.
- In order to be considered accurate, the model should agree with the calculated results to within 10%.
- Takes into account:
 - Wave attenuation
 - Reflections/refractions at boundaries



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.

L1 Requirement – 1.2

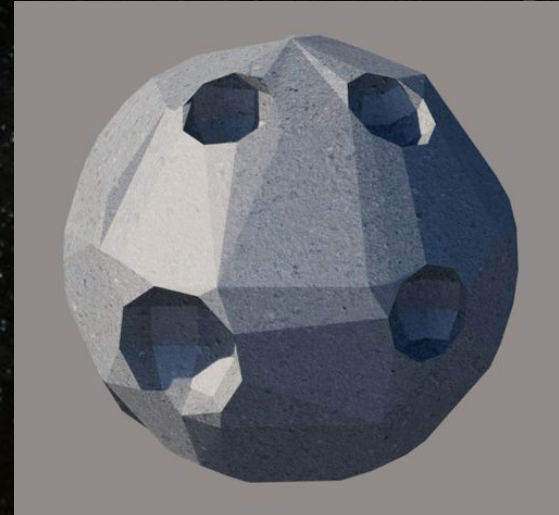
- Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)
- Considerations:
 - Hardware (accessibility, performance, compatibility with software)
 - Software (CPU architecture, package management, Operating System, file I/O)



Corredor, John & Rodriguez, Camilo. (2021). Beowulf Cluster to support disciplinary education and research in Computer Science. 10.13140/RG.2.2.36786.12481.

L1 Requirement – 1.3

- Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments and simulated asteroid.
- Includes:
 - Varying Model Characteristics:
 - Density
 - Porosity
 - Layering
 - Varying Inputs:
 - Initial Energy
 - Locations of “Impactor” and “Detector”
 - “Impact” angle (initial direction of waves)



Romanenko, K. (2019, June 17). *asteroid-5*.
GrabCAD.

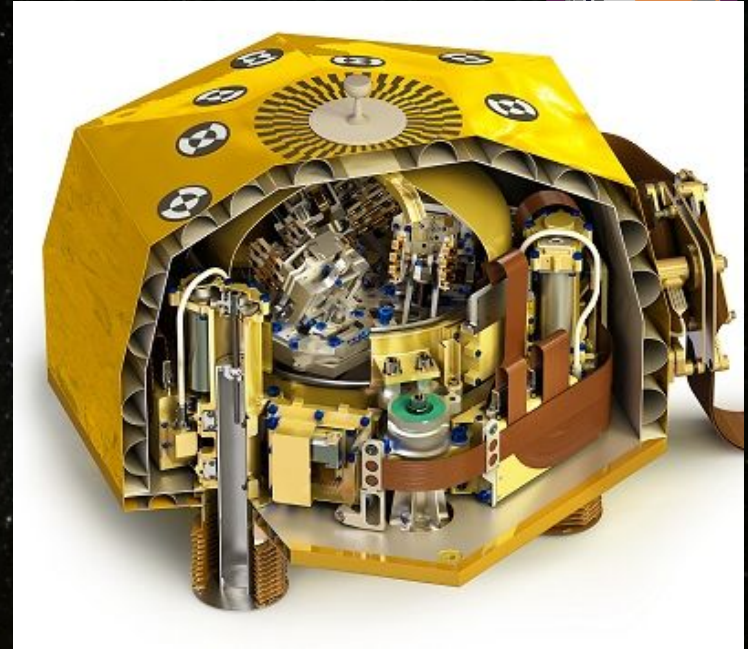
Subsystem Requirements - 1.1



ID	Requirement	Verification	Parent	Rationale
L2-WAVE-1	Basic tests of the simulation code shall use homogeneous spheres of various densities approximating 16 Psyche - a radius of 115 km	Test, Analysis	L1.1	Confirms the accuracy of the simulation code
L2-WAVE-2	The readings at the simulated detector shall be within the sensitivity limits of the SEIS instrument, with a ground displacement of 1 Å and a ground acceleration spectral density of $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$	Demonstration, Analysis	L1.1	Confirms the waves could be detected by a real-world seismometer

Science Subsystem

- Accuracy - The simulation code's accuracy will be confirmed by comparison to manually calculated results in simple examples
- Measurability - The simulated seismic waves at the location of the detector must be above the noise floor of real-world detectors
 - SEIS - The SEIS (Seismic Experiment for Internal Structure) is a seismometer on Mars included with the InSight Mission - due to its proven use on a non-Earth object it will be our real-world reference point
 - The SEIS is sensitive to a ground displacement of $1 \text{ \AA}$ and ground acceleration spectral density of $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$



Interactive VBB pendulum. L'instrument seis - SEIS / Mars InSight. (2018, February 26).

Science Subsystem - Legacy Equipment



- In order to improve the feasibility of the concept, we plan to simulate legacy equipment whenever possible
 - This includes the SEIS
- The minimum impactor kinetic energy we estimated could be detected on opposite sides of 16 Psyche is approximately 3.8 MJ
 - JAXA's Hayabusa2 mission used the Small Carry-on Impactor (SCI) which had a kinetic energy of 4 MJ - another piece of legacy equipment available for this concept

$$E_I = \frac{E_f r^2}{k} e^{-\alpha}$$

Impact Kinetic Energy Formula

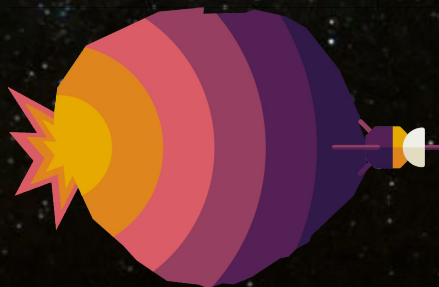
Where

- E_I = kinetic energy of impactor
- E_f = wave energy at location of detector
- r = distance between impact site and detector
- α = decay constant
- k = seismic efficiency of impact

Science Subsystem - Question Answers

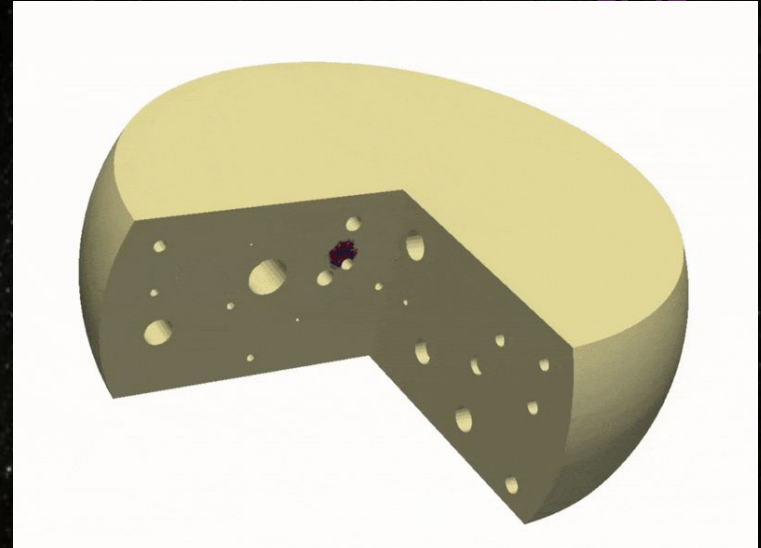


- Signal strength
 - What is the ground acceleration at the location of the detector
- Comparison with noise floor
 - If the ground acceleration is equal to or greater than the minimum detection threshold we have set (based on the SEIS) the science question will be answered “yes”
 - If it is less, then the answer will be “no”
- Length of signal
 - While not strictly required to show feasibility, the greater the length of the seismic signal, the easier it is to both detect and use for interior mapping



Model Subsystem

- To begin any simulation, SeisSol needs to internally construct a model, which can be accomplished via two methods:
 - Importing a mesh, specifically in the .msh file format
 - Mathematically defining a basic shape in a YAML parameter file
- A mesh is generated from an original CAD model using 3D tetrahedral elements
 - End goal number of mesh elements depends on the final hardware



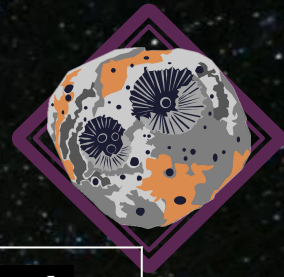
Video Credit: Thomas Ulrich - 2019

Model Subsystem Interfaces



- Input Interfaces
 - SeisSol Software
 - Dependencies
 - SimModeler Software
 - Dependencies
 - Parameters File (.spj)
 - Defines shape of model, size, layering, etc.
- Output Interfaces
 - 3D CAD Model (.stl)
 - Internal Surface File (.msh)
 - Simulation Output Files (.h5)

Subsystem Requirements - 1.2



ID	Requirement	Verification	Parent	Rationale
L2-COMP-1	The OS and other dependency software shall support the successful operation of SeisSol	Test	L1.2	Prevents compilation and runtime errors
L2-COMP-2	The hardware models and setup shall meet the minimum performance required to successfully complete a run of SeisSol lasting no longer than 24 hours	Demonstration	L1.2	Allows for multiple runs of various 16 Psyche Models

Computer Subsystem - Software



- SeisSol - Our primary seismic wave simulation software.
- Operating System - SeisSol was developed on a unix-based OS. For maximum compatibility, we should run it on the same OS.
- Package Management - SeisSol utilizes Spack Open Source Package Manager for installation of the program, as well as all of its dependencies.
- CPU Architecture - Host architecture and corresponding command is needed for compiler tuning variables and utilization of proper code generators upon build phase, allowing for optimal build performance (TBD by hardware specs)

Computer Subsystem - Hardware



- CPU - Performance depends on multiple factors:
 - **Individual CPU performance:** Clock speed, number of cores, instructions per clock (IPC), number of threads for multitasking, etc (TBD)
 - **Number of CPUs in Setup:** One CPU vs a CPU cluster system vs ASU's Sol supercomputer. (TBD)
- Memory - Random Access Memory (RAM). Performance reliant on:
 - **Number of Gigabytes** - There is a minimum amount of RAM needed to meet our runtime demands (TBD)
 - **DDR (Double Data Rate)** - DDR Generation may matter more with less RAM, and less with more RAM (TBD)

Computer Interfaces



- Input Interfaces
 - Environment
 - Temperature control
 - Minimal contaminants (dust, moisture)
 - Stable and mount or stand to ensure no interruptions to or damaging of hardware.
 - Power draw from lab (operation, and performance)
 - Keyboard, mouse - interact with OS via user interface (UI)
 - UI - interact with and instruct software to perform operations and run simulation
- Output Interfaces
 - USB, USB-C, DP, HDMI or some other streaming methodology that properly displays the UI on a screen and, by extension, the results of the simulation.

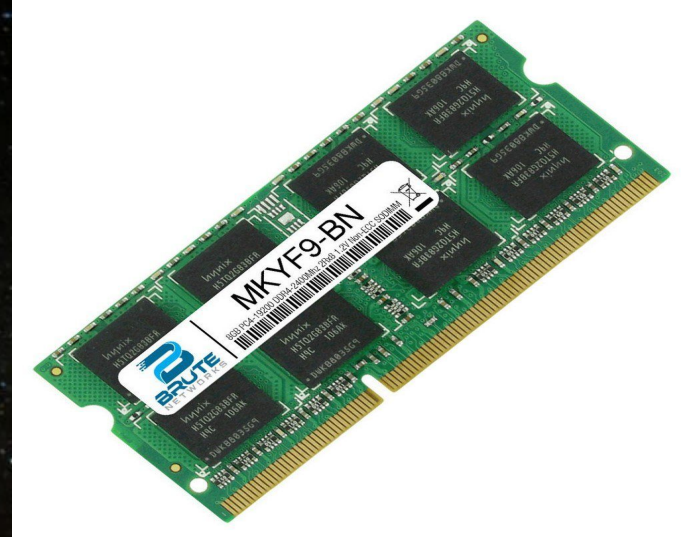
CPU

- Intel Core i7-7600u, laptop model
 - Thermals may be suboptimal for performance in this casing
- 2 cores
- 4 threads
- Base Frequency - 2.80 GHz
- Boost Frequency - 3.90 GHz
- Max Supported Memory - 32 GB
- Supported Memory Bandwidth - 34.1 GB/s
- Supported Memory Types (for maximum bandwidth above) - DDR4-2133, LPDDR3-1866, or DDR3L-1600



RAM

- 2, 8GB RAM Sticks (part #: MKYF9)
- 16GB Total of RAM
- DDR4 - second best double data rate generation
- Clock speed (frequency) - 2400 MHz
- Current test runs show more RAM is needed (sim is exiting due to running out)
- Could upgrade system to 32 GB of RAM (slots and budget permitting)



Software Key Components



- OS
 - Any recent, stable Linux distribution will work.
 - Current distribution - Ubuntu 22.04.5 LTS
- Package Management
 - Linux has built in package managers
 - Official recommendation in documentation for SeisSol is SPACK
- CPU architecture
 - The Intel Core i7-7600u is based on the Kaby Lake architecture
 - SeisSol does not support Kaby Lake specifically, but does support other i7 architectures
 - Current SeisSol build was compiled with the Intel Haswell architecture
 - “HOST_ARCH = hsw”

Computer Testing



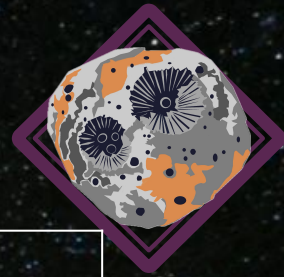
- **Test COMP-01: SeisSol Operational Test**

- Purpose: Ensure that SeisSol was properly installed and compiled
- Test: Run the following command line prompt, directed at the proper installation folder:
 - `~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic --help`
- Pass/Fail
 - Pass: Return message consisting of a welcome message, copyright, version, build date, last commit ID, and some other qualifiers
 - Fail: No terminal reply, a “no such file or directory” bash message, or any other error message

- **Test COMP-02: Hardware Performance Test**

- Purpose: Ensure that the runtime does not exceed 24 hours for the minimum model complexity that is acceptable in answering our science question
- Test: Iterative. Start with a test model (100-200 mesh elements), then increase complexity in subsequent runs up to the accepted level of complexity. Keep track of runtime for each iteration.
- Pass/Fail
 - Pass: All iterations exit runs in 24 hours or less
 - Fail: Any of the iterations take longer than 24 hours

Subsystem Requirements - 1.3



ID	Requirement	Verification	Parent	Rationale
L2-SIM-1	The simulation code shall be able to support multiple receivers and varying positions	Inspection	L1.3	Shows effect of varying receivers on data
L2-SIM-2	The simulation code shall be able to support varying impact angles, positions, and energies.	Inspection	L1.3	Shows effect of varying impacts on data
L2-SIM-3	The simulation code shall be able to support varying density, layering, impact, energy efficiency, and porosity of 16 Psyche models	Inspection	L1.3	Shows effect of varying conditions of 16 Psyche on data

Simulation Subsystem

- Custom CAD importing and meshing
- Simulate point source impact
- Allow for control over initial conditions (material density, impact source point, wave velocities)
- Visualize wave pathing through 3D Psyche model
- Output total energy of simulation



Simulation Subsystem Interfaces



- Input Interfaces
 - Initial Conditions Parameter File
 - Model density and layering
 - Wave velocities and energy
 - Seismic source type
 - Model Mesh
 - Input mesh map for Psyche model
- Output Interfaces
 - Wavefield data
 - Visualization metadata files for use in Paraview

Simulation Subsystem Testing



- **Test SIM-01: SeisSol Initial Conditions**
 - Purpose: Determine level of control for desired initial conditions
 - Test: Write an “easi” file script using the “easi” file directory to control desired initial conditions and mesh
 - Pass: “easi” parameter file scripted to control number of seismic receivers on our mapped model and their positions; point-source seismic origin; model density, layering, seismic magnitude, energy efficiency, and model porosity
- **Test SIM-02: Simple Geometry/Mesh**
 - Purpose: Use imported spherical model of 16 Psyche and apply different meshes and conditions to run simulations
 - Test: Open a terminal, `cd ~/my_simulationSimple`, then:
`mpirun-np1~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic parameters.par`
 - Pass: Saved wavefield output file and surface output metadata files for visualization in Paraview, detectable readings

Simulation Subsystem Testing Cont.



- **Test SIM-03: Complex Geometry/Mesh**
 - Purpose: Use imported similar model of 16 Psyche and apply different meshes and conditions to run simulations
 - Test: Open a terminal, cd ~/my_simulationComplex, then:
mpirun-np1~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic parameters.par
 - Pass: Saved wavefield output file and surface output metadata files for visualization in Paraview, detectable readings

Testing Subsystem



- To confirm our results, we can compare the the energy from our test simulation with our calculations.
- The Seismic wave energies can be calculated by using the initial values by hand
- By using the values used for:
 - v_p = longitudinal wave velocity
 - v_s = transverse wave velocity
 - F = momentum of impactor
 - τ = time constant
 - ρ = density
- The total Seismic Wave Energy ($v_s + v_p$) equation can be separated into energies for the longitudinal and transverse waves

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Testing Subsystem Interfaces



Only Simulation data and calculations, no external input

Inputs:

- Longitudinal seismic-wave speed (v_p)
- Shear Wave speed (v_s)
- Density
- Impactor momentum
- Time constant
- Model layers and geometry
- Impactor and detector locations and distance
- simulated detector energy + arrival times.

Outputs:

- ideal energy calculation,
- % difference between simulated and calculated values
- pass/fail based on requirements.

Testing Key Components



Energy Calculation:

- Manually calculate ideal energy using equation.
- Use same initial values as the simulation

Result comparison:

- Calculate the value difference between ideal and simulated energy.
- Must be within 10% to count as accurate.

Minimum Detection Threshold:

- Detector must receive more than 4 MJ

Wave Behavior

- Compare the timing and speed of waves (v_s and v_p) in simulation to the calculated expectation
- Should be within 10% to be verified.

Testing Plan



Test 1 - Energy Accuracy

- Compare simulation energy with calculated ideal value.
- Must be within 10%.
- If not, the run fails.

Test 2 - Minimum Energy Required

- Check if detector gets at least 4 MJ.
- If not, the run fails.

Test 3 - Wave Timing

- Compare simulated wave arrival time with expected travel time.
- Time should be close to the calculated value (10%).

List of Tracked Risks



ID	Summary	L	C	Trend	Approach	Risk Statement
1	Power	1	3	NEW	R	The system loses power during runtime
2	Budget	1	5	NEW	W	The project exceeds a budget of \$2000
3	Hardware	2	4	NEW	M	Computer suffers major component failure
4	Schedule	2	5	NEW	M	Unexpected delays (installation, user error)
5	Software	1	3	NEW	R	The software suffers a runtime failure or corruption
6	Facilities	1	3	NEW	R	Our group cannot get access to facilities to base out of
7	Access	5	4	NEW	M	Can't get access to higher performance hardware if needed

L = Likelihood (1-5)

1 = not likely

5 = extremely likely

C = Consequence (1-5)

1 = low consequence

5 = high consequence

LxC Trend

↓ - Decreasing (improving)

↑ - increasing (worsening)

→ - unchanged

NEW - added this week

Approach

A - accept

M - mitigate

W - watch

R - research

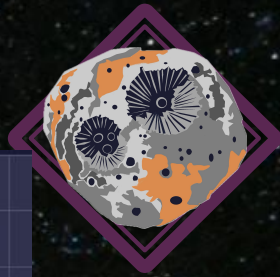
Risk Matrix



Criticality
HIGH
MED
LOW

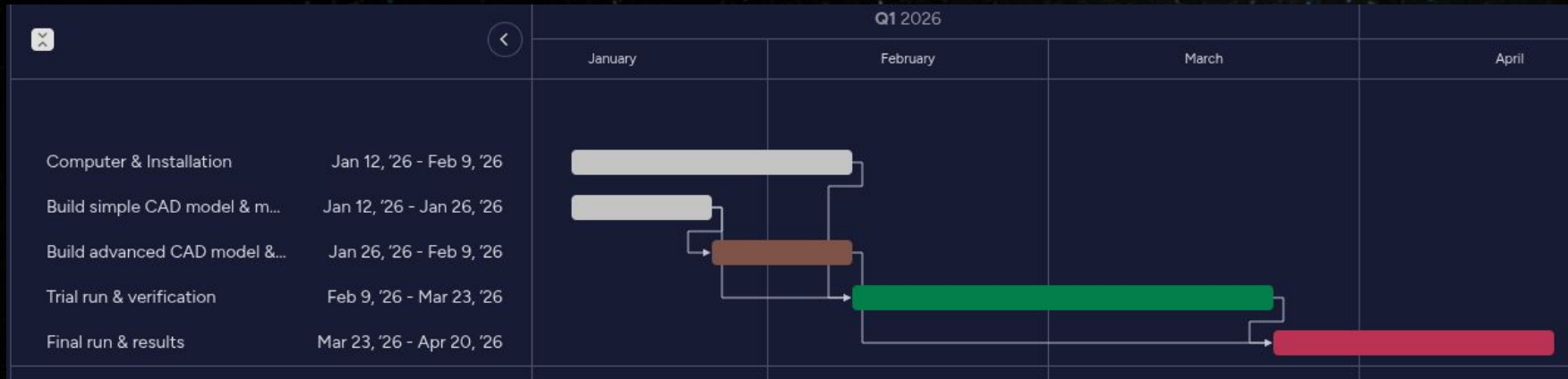
L I K E L I H O O D	5					
	4					7
	3					
	2				3	4
	1			1, 5, 6		2
		1	2	3	4	5
CONSEQUENCES						

Schedule



☐	Task		Status ⓘ	Timeline ⓘ	
☐	Computer & Installation		✦ ⊕	Working on it	Jan 12, '26 - Feb 9, '26
☐	Build simple CAD model & mesh		✦ ⊕	Working on it	Jan 12, '26 - Jan 26, '26
☐	Build advanced CAD model & mesh		✦ ⊕	Not Started	Jan 26, '26 - Feb 9, '26
☐	▼ Trial run & verification 2		✦ ⊕	Not Started	Feb 9, '26 - Mar 23, '26
☐	Subitem		Status	Date	+
☐	Debugging & running	⊕		Mar 9, 2026	
☐	Verifying output results	⊕		Mar 23, 2026	
☐	+ Add subitem				
☐	▼ Final run & results 2		✦ ⊕	Not Started	Mar 23, '26 - Apr 20, '26
☐	Subitem		Status	Date	+
☐	Debugging & running	⊕		Apr 6, 2026	
☐	Output results	⊕		Apr 20, 2026	

Schedule – Gantt Chart



Budget



- RAM/Memory improvements
 - Current hardware: 16 GB
 - Possible necessary RAM: 64-128 GB
- Minimum average of 16 & 32 GB DDR5 RAM sticks, 4 slots for standard motherboard
- High price fluctuations
- Current Estimated Cost:
 - Minimum \$500 for 64 GB
 - Minimum of \$1000 for 128 GB
- Arizona State University offer access to computers (including the Sol supercomputer) for free for all students.
- Current selections for simulations are open source

Conclusions & Next Steps



- Our science question depends on a simulation of probable internal conditions and structural layers of 16-Psyche, and necessitate the use of SeisSol to produce a reasonable map of 16-Psyche's interior.
- Next steps:
 - Follow through on CDR notes and begin preparation for next semester.
 - Continue working towards gaining access to system that can support SeisSol.
 - Explore alternatives
 - SeisSol is currently our chosen software, but having backup options will mitigate risk.



**Special thanks to our mentor, Cassie Bowman,
as well as ASU experts Samuel Courville and
Edward Garnero**

**Thank
You!
Questions?**



Technical Readiness Level (?)



Backup Slides:



Requirements Must be SMART:

- Specific -
 - It must address only one aspect of the system design or performance
 - It must be expressed in terms of the need (what and how well), not the solution (how).
- Measurable -
 - Performance is expressed objectively and quantitatively
 - For example, an exact pointing requirement (in degrees) can be tested thus verified prior to launch.
- Achievable -
 - It must be technically achievable at costs considered affordable
 - Note: JWST early designs specified an aperture requirement eventually descope due to technical issues with deployment.
- Relevant -
 - It must be appropriate for the level being specified
 - For example, requirement on the solar cells should not be designated at the spacecraft level.
- Traceable -
 - Lower level requirements (children) must clearly flow from and support higher level requirements (parents).
 - Requirements without a parent are referred to as orphans, and need to be assessed for necessity of inclusion.

STM Contents

(Groppi)

- Teams developing NASA missions use an STM to summarize the science and top-level requirements.
- An STM has the following columns:
 - **Science Objectives:** The elements of your science question(s)
 - **Measurement Objectives:** What quantities do you need to measure to achieve the science objective:
 - **Science Measurement Requirements:** What do you need to measure and how well do you need to measure it? Expected measurement range and required measurement error.
 - **Instrument Functional Requirements:** What other requirements does the instrument that takes the data need to have to meet your needs?
 - **System Functional Requirements:** What other requirements does the whole experiment need to have to meet your requirements?
 - **Science Product:** What is the data your experiment will produce (the low-level data measured, not the answer you get after you analyze it).

Basic simulation

It consists of:

- 1 Impact point
- 1 detection point
- 2 -3 layers

An example of SeisSol simulation for 2 layers

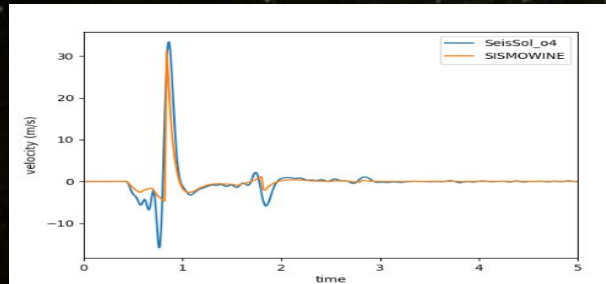
From central point (0,0,0)

Layer: from 0 to 1000 (z direction);

Half-space: 1000 to 3300 (z direction)

Impact source: (0, 0, 2000).

We can pick any impact and detection point by specifying the coordinates,

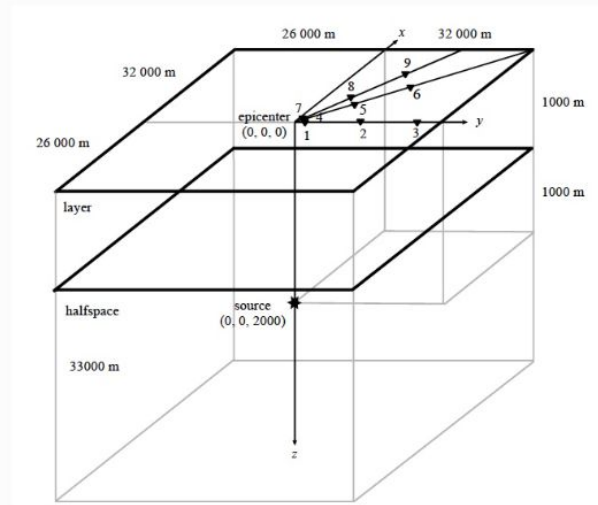


Example of a basic simulation of LOH1



	Vp (m/s)	Vs(m/s)	density	Qp	Qs
layer	4000	2000	2600	Inf	Inf
half-space	6000	3464	2700	Inf	Inf

Table: Material properties in LOH1.



Geometry of LOH1.

Requirements



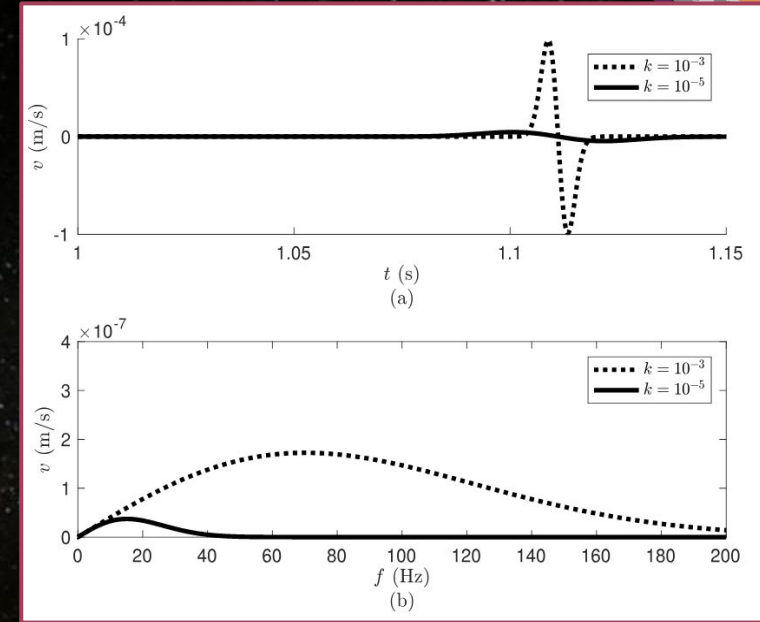
Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
MR1.1	Propagation of seismic waves - The simulation shall to accurately model the propagation of seismic waves through an artificial 3d body.	Output of test model matches manually calculated tests to 1 Order of Magnitude	Science Question
MR1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	The simulation completes 1 run in 24 hours or less	Science Question
MR1.3	Initial Conditions - The simulation code must be able to support varying initial conditions within the simulated instruments and simulated asteroid.	The simulation code can be run repeatedly with varying initial conditions	Science Question

Mission Requirement – 1.1



- Use previously acquired data to estimate frequency range, needed sampling rate, and timing accuracy.
- Use finite element models and seismic wave propagation simulations of Psyche's interior to refine previously estimated data.
- Use simulations to determine optimal seismometer placement.
- Measure changes in seismic signal characteristics as they propagate to map the subsurface interior.



Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.2



- Impact device shall generate enough energy to propagate seismic signal throughout the entirety of 16 Psyche.
- Requires multiple preliminary assumptions that are not yet known to be accurate to 16 Psyche.
- Simulation will be needed to better estimate 16 Psyche's wave velocities, seismic efficiency of impact, $k = E_s / E_k$, and other key values.

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

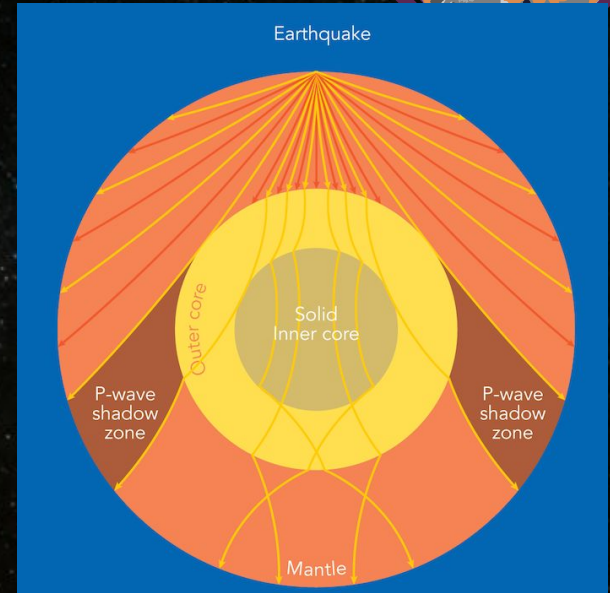
Where

- F = momentum of impactor
- v_p = longitudinal wave velocity
- v_s = transverse wave velocity
- τ = time constant
- ρ = density of target surface

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.3

- The simulation shall be able to account for varying numbers and positions of impactors and/or detectors.
- More impactors/detectors will improve precision of measurements.
- But larger numbers will also be less practical.



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.